DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Safety & Health

CHECK THE TAG.

SCAFFOLDS, SWING STAGES AND AERIAL LIFTS PUT YOU UP HIGH ON SOMETHING SOMEONE ELSE BUILT. KNOW WHO CHECKED IT AND HOW YOU GET DOWN.

WHO BUILDS IT

ONLY THE SCAFFOLD CREW BUILDS, CHANGES OR TAKES DOWN A SCAFFOLD.

A trained scaffold crew erects, moves, alters and dismantles every scaffold, under a **competent person** named by the foreman. The competent person inspects it **before every shift** and after any sandstorm, wind that cleared it, dust devil, impact or change, then signs the tag. Need a plank moved or a rail off? Ask your foreman. **Never do it yourself.**

4×	10 ft	Every shift
ITS RATED LOAD, BUILT IN	FALL PROTECTION ON SCAFFOLDS	COMPETENT PERSON INSPECTS

Before you work on a scaffold, swing stage or lift: this supplement on your badge. Before you build scaffolds or run a lift: the scaffold erector or MEWP operator endorsement as well. Scaffolds taller than 125 ft, cantilevered, carrying screens or netting, or carrying unusual loads are designed by Design 0 Structures.

SCAFFOLD TAGS

SITE RULE · A TAG AT EVERY ACCESS POINT

GREEN
SAFE TO USE
Inspected this shift. Shows the competent person's name, the date, the shift and the load rating. Normal rules apply.

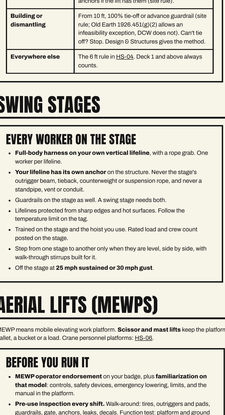
YELLOW
USE WITH CARE
Usable only under the conditions written on the tag: harness and lanyard required, a rail out for loading, a reduced load. Read it every time.

RED
DO NOT USE
Being built, changed or taken down, failed inspection, or damaged. Only the scaffold crew goes on it.

No tag, or a tag not signed this shift, means red (site rule). Old Earth requires the inspection. DCW adds the tag.

A SOUND SCAFFOLD

OSHA 1926.451



- **Footing:** base plates on mud sills, on compacted, level ground. Never on loose sand, blocks, bricks, drums or pallets. Wind scours sand out from under sills, so they are checked after every blow.
- **Plumb, level and braced.** Screw jacks within the maker's limit.
- **Ties:** taller than 4 times its base width? It is tied, guyed or braced to the structure. Ties no more than 20 ft apart going up (26 ft if the scaffold is wider than 3 ft) and 30 ft apart along it. Only the scaffold crew removes a tie.
- **Platforms:** fully planked or decked, gaps 1 in or less, at least 18 in wide. Planks cleated, or run 6–12 in past their supports. Every plank secured against wind (site rule).
- **Guardrails** on every open side and end: top rail 38–45 in, midrail halfway, toeboard at least 3.5 in. Top rail holds 200 lb (100 lb on a swing stage).
- **Access:** ladder, stair tower or ramp wherever the platform is more than 2 ft above or below where you step on. Never climb cross-braces or guardrails.
- **Load:** built to hold 4 times its rated load. That margin is not yours to use. The tag shows the rating: light duty 25 lb/sq ft (people, hand tools), medium 50 (stacked block, panels), heavy 75 (masonry, heavy material).
- **Power lines:** 10 ft minimum up to 50 kV, more above (Supplement B (HS-SB)).

FALL PROTECTION

WHERE YOU ARE	WHAT PROTECTS YOU
Scaffold platform 10 ft or more above a lower level	Guardrails on every open side, or a personal fall arrest system
Scaffold beside an open deck edge	Count the fall to the level below that edge
Swing stage	Guardrails **and** your own vertical lifeline. Both, always.
Boom lift, any height	Full-body harness and restraint lanyard on the maker's anchor in the platform
Scissor or mast lift	Guardrails and gate closed. Clip in to the maker's anchors if the lift has them (site rule).
Building or dismantling	From 10 ft, 100% tie-off or advance guardrail (site rule; Old Earth 1926.451(g)(2) allows an infeasibility exception, DCW does not). Can't tie off? Stop. Design 0 Structures gives the method.
Everywhere else	The 6 ft rule in HS-06. Deck 1 and above always counts.

DO

- Check the tag at the access every time you go up.
- Sweep sand and debris off planks. Sand on a plank is like marbles.
- Tool lanyards on hand tools. Keep material back from the edge.
- Clip in whenever the tag says so.
- Lock rolling-scaffold casters before anyone climbs.
- Report a missing plank, loose tie or damaged part at once.

DON'T

- Climb cross-braces or guardrails.
- Stand on a ladder, bucket or box on the platform.
- Remove or move a plank, rail, brace or tie.
- Ride a rolling scaffold while it moves (site rule).
- Hang tarps, shade cloth, netting or signs. They turn a scaffold into a sail. Only the scaffold crew fits them, to a Design 0 Structures design.
- Throw anything down. Lower it.

SWING STAGES

TWO-POINT SUSPENSION SCAFFOLDS · OSHA 1926.451(D), (G)

EVERY WORKER ON THE STAGE

- **Full-body harness on your own vertical lifeline**, with a rope grab. One worker per lifeline.
- **Your lifeline has its own anchor** on the structure. Never the stage's outrigger beam, tieback, counterweight or suspension rope, and never a standpipe, vent or conduit.
- Guardrails on the stage as well. A swing stage needs both.
- Lifelines protected from sharp edges and hot surfaces. Follow the temperature limit on the tag.
- Trained on the stage's and the hoist you use. Rated load and crew count posted on the stage.
- Step from one stage to another only when they are level, side by side, with walk-through stirrups built for it.
- Off the stage at **25 mph sustained or 30 mph gust**.

RIGGING, EVERY SHIFT

The competent person checks the rigging before every shift and after anything that could damage it: impact, heat, a storm.

- Outrigger beams, tiebacks and anchors match the rigging plan. Tiebacks run to a sound structural anchor.
- **Counterweights:** only the maker's counterweights, fixed to the beam. **Never sand, sandbags, water or loose material.**
- Suspension wire rope: no broken wires, kinks, birdcaging, crushing, corrosion or heat damage. Fiber-core wire rope is never used above 180°F.
- Hoists tested and tagged. Brakes hold. A slipping brake stops the job.
- Stage tied in to the structure so it can't swing, as the competent person sets.
- Platform no wider than 36 in unless Design 0 Structures designs it.
- The ground below is barricaded.

AERIAL LIFTS (MEWPS)

ANSI/SAIA A92.22 SAFE USE · A92.24 TRAINING

MEWP means mobile elevating work platform. **Scissor and mast lifts** keep the platform over the base. **Boom lifts** reach out past it, so a jolt can throw you out. Never ride a forklift pallet, a bucket or a load. Crane personnel platforms: HS-06.

BEFORE YOU RUN IT

- **MEWP operator endorsement** on your badge, plus **familiarization on that model**: controls, safety devices, emergency lowering, limits, and the manual in the platform.
- **Pre-use inspection every shift.** Walk-around: tires, outriggers and pads, guardrails, gate, anchors, leaks, decals. Function test: platform and ground controls, emergency stop, emergency lowering, tilt and descent alarms, horn, anti-entrapment device.
- Anything wrong: tag it out, park it, tell your foreman.
- Check the work spot: slopes, holes, edges, trenches, soft sand, fresh backfill, overhead structure, power lines, traffic, wind.
- **A trained ground person stays at the base** while the platform is up. They know the ground controls and the emergency lowering, keep traffic out, and are your buddy (site rule).
- Everyone in the platform knows the emergency stop and how to lower it.
- Battery life charge only in charging bays (HS-SB).

GROUND & SAND

- **Firm, level ground** within the maker's slope limit. Loose sand won't hold a wheel or an outrigger.
- Every outrigger or stabilizer on a pad or mat sized by the maker. Never on fresh backfill or near a trench edge.
- **Tilt alarm sounds:** stop, lower the platform, set up again.
- Drive with the platform raised only where the maker allows it, on firm, smooth, level ground. On sand, slopes and rough ground, lower it first.
- Wheels chocked on any slope.
- **Spotter** wherever you drive, work near edges or overhead structure, or work in traffic. The spotter stays in your view. Barricade the base.

IN THE PLATFORM

- **Boom lift:** full-body harness, with a restraint lanyard clipped to the maker's anchor in the platform **before it leaves the ground**. Use the lanyard type the maker names.
- **Never clip to the structure**, steel or a beam next to you. If the lift moves, it pulls you out.
- **Scissor lift:** rails in place, gate or chain closed. Clip in to the maker's anchors if the lift has them (site rule).
- Feet on the floor. Never stand, sit or climb on the rails, or use a ladder, bucket or box in the platform.
- The rated load counts people, tools, material and water. Crew count is on the plate.
- No sheet goods, panels or signs on the platform unless the maker allows it. They act like a sail.
- **Never leave the platform at height** unless the maker allows it, your foreman has a written plan, and you stay tied off 100% (site rule).

OVERHEAD & CRUSH

- **Look up** before you raise, swing or drive. Slow down near overhead steel, decks and ducts.
- Keep your head and body inside the rails. Never lean over the controls.
- Never disable the anti-entrapment device or any interlock.
- **Trapped against structure?** Hit the emergency stop. The ground person reverses the last move from the ground controls only if they are sure it frees you. Ch 1 "Caught-in".
- **Power lines:** 10 ft minimum up to 50 kV, more above (HS-SB). Lines sway in wind.
- Never use a lift as a crane, or to push or pull.

WIND & WEATHER

SITE WIND TABLE (SITE RULE) · CH 16 ATMOS

SITE WIND TABLE

Wind = the higher of the local anemometer (crane, MEWP or crew meter) and the Ch 16 posting. Gust = 3-second gust.

WIND	RULE
15+ mph	Goggles outside cabs. Secure sheet goods and loose material.
20+ mph sustained or gust	No outdoor hot work. No crane personnel platforms. High-sail loads need Lift Director approval.
25 mph sustained or 30 mph gust	**Off open steel, roofs, leading edges, swing stages and outdoor scaffolds above Dock 1. MEWPs down (or the maker's limit if lower).**
30 mph sustained or gust	All crane loads landed; booms secured per maker.
Blowing-dust advisory (visibility 1 mile or less)	No outdoor hot work. Lifts stop whenever the operator loses sight of the load or signal person.

The MEWP maker's limit is on the platform plate, often 28 mph. A lift rated 0 mph is indoor-only and never goes outside. Wind is stronger at height: read the meter at platform level.

STAGED WARNINGS

- **Sandstorm watch** (storm possible within 2 hours): secure loose material, planks and tarps on every platform. Plan how each platform comes down. Stage crews near shelter.
- **Sandstorm warning** (arrival within 30 minutes): MEWPs down and parked, everyone off swing stages and scaffolds, to shelter. The take-shelter siren sounds no later than 15 minutes before arrival.
- **Lightning alert** (strikes within 10 miles): out of MEWPs, off scaffolds and steel, into sealed shelters or closed vehicles, never under steel. Resume 30 minutes after the last strike within 10 miles.
- **After any storm**, any wind that cleared the scaffold, or a dust devil through it: the competent person re-inspects and re-tags before anyone goes up.
- **Never restart on a siren tone alone.** Wait for the Ch 1 voice all clear or your crew lead.

STUCK AT HEIGHT

A lift dies with the platform up, a worker hangs in a harness, or someone up top is hurt or overheating.

NO RESCUE PLAN, NO LIFT.

Before anyone goes up, your crew lead names the rescue method, the trained ground person and the backup, and the competent person checks the plan (site rule). The plan is written and posted at the job. Waiting for Fire & Rescue takes too long for a worker hanging in a harness.

1. **Someone hurt, ill or hanging?** Ch 1: "Emergency, emergency, emergency" with the type ("Fall, worker suspended", "Heat casualty", "Man down"), location, number hurt, hazards and your call sign. Stay on Ch 1 until released.
2. **Lift won't move:** the person in the platform tries the platform controls, then auxiliary power.
3. **The ground person brings it down** from the ground controls, then with the emergency lowering marked on the machine.
4. **Swing stage stuck:** workers stay on the stage, clipped to their lifelines. The competent person runs the stage rescue plan.
5. **Worker hanging in a harness:** run the fall rescue now (HS-04). Suspension trauma can make them pass out within minutes.
6. **Never climb out** along a boom or onto structure to get down unless rescue directs it.
7. **Stuck but well:** stay clipped in and tell your foreman. In the heat, any sign of heat illness means Ch 1 "Heat casualty".

HEAT ON PLATFORMS

- Sun-baked platform floors, rails and scaffold tubes burn skin. Gloves on. Kneel on a pad.
- A harness traps heat. Climbing counts as heavy work on the work/rest table (HS-03). In FR leathers or a sealed coverall, add the clothing adjustment first. RAMP band on? Half the table's work minutes on Days 1–4.
- Water goes up with you in a tethered bottle. It counts toward the platform load.
- **No lone work at height.** Your buddy is on the platform or is the ground person. From Extreme caution up, buddies stay within sight of each other.
- Shade umbrellas and sails catch wind. MEWPs: only if the maker allows it. Scaffolds: only the scaffold crew fits them, to a Design 0 Structures design. Never fit one yourself.
- **Dizzy, cramping or confused?** Sit down, stay clipped in, lower the platform and call your buddy. Anyone not fully alert is a heat-stroke case: Ch 1 "Heat casualty".

STOP-WORK AUTHORITY

Anyone can stop unsafe work. No penalty, no questions. Stop it, make it safe, tell your crew lead (your foreman if there's no crew lead; any green hat if it isn't fixed). Work restarts only when the crew lead confirms the fix with the person who stopped it. Every stop is logged.

REPORT IT

A damaged scaffold, a tipped or stranded lift, a dropped tool: all of it gets reported. Report every injury, illness, first aid case, near miss, damage and unsafe condition to your crew lead before the end of the shift (immediately for anything serious), on form HS-IR with your Colonist ID. Don't move anything except to rescue. Safety & Health investigates with the crew; findings are shared at muster. No retaliation.

WORKER ACKNOWLEDGEMENT

I will check the scaffold tag every time, never change a scaffold myself, clip in where this poster says, inspect every lift before I use it, and never go up without a rescue plan.

NAME _____ COLONIST ID _____ CREW / SECTOR _____ DATE _____

Adapted from Old Earth standards: OSHA 29 CFR 1926 Subpart L (1926.451 scaffold capacity, platforms, footing, ties, access, use, inspection, power-line clearance, fall protection and suspension scaffolds; 1926.452 specific scaffolds (wc), two-point suspension and mobile scaffolds; 1926.453 aerial lifts; 1926.454 training; Appendix A duty ratings), OSHA 1926.502 (fall protection systems), ANSI/SAIA A92.20, A92.22 and A92.24 (MEWP design, safe use and training), ANSI/ASSP A10.8 (scaffolding). Wind table, scaffold tags and items marked (site rule) are DCW site rules.